

Polynomial Rings Construction

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1 Construction: Polynomial Ring in a set of variables

Let R be an arbitrary ring.

Condition: Let R be a ring, X be a nonempty set.

Construct the set of monomials:

$$\mathcal{M} := \{\alpha \in \omega^X \mid \exists(n \in \omega) \ n = |\alpha^{-1}(\omega - \{0_\omega\})|\}$$

Construct polynomials:

$$R[X] := \{f \in R^\mathcal{M} \mid \exists(n \in \omega) \ n = |f^{-1}(R - \{0_R\})|\}$$

Construct addition:

$$+_R[X] := \{\langle \langle f, g \rangle, h \rangle \in R[X]^2 \times R[X] \mid \forall(m \in \mathcal{M}) \ h(m) = f(m) +_R g(m)\}$$

Construct multiplication:

$$\times_{R[X]} := \{\langle \langle f, g \rangle, h \rangle \in R[X]^2 \times R[X] \mid \forall(\gamma \in \mathcal{M}) \ , \ h(\gamma) = \sum_{(\alpha, \beta) \in D_\gamma} f(\alpha) \cdot_R g(\beta)\}$$

where $D_\gamma := \{\langle \alpha, \beta \rangle \in \mathcal{M}^2 \mid \alpha \in \text{supp}(f) \wedge \beta \in \text{supp}(g) \wedge (\alpha +_{\omega^X} \beta = \gamma)\}$

Edge case: we define $R[\emptyset] := R$

Theorem 1.1. (polynomial rings of two sets of variables)

Let R be a ring, X, Y be two nonempty sets. Then,

$$R[X][Y] \cong R[X \sqcup Y]$$

Proof. Let

$$\mathcal{M}_X := \{\alpha : X \rightarrow \mathbb{N} \mid \alpha \text{ has finite support}\},$$

and similarly define $\mathcal{M}_Y$ and $\mathcal{M}_{X \sqcup Y}$.

We first define a bijection

$$\Psi : \mathcal{M}_X \times \mathcal{M}_Y \rightarrow \mathcal{M}_{X \sqcup Y}$$

by

$$\Psi(\alpha, \beta)(z) = \begin{cases} \alpha(z), & z \in X, \\ \beta(z), & z \in Y. \end{cases}$$

This is well-defined and bijective, with inverse given by restriction to X and Y .

Definition of the map. Let $f \in R[X][Y]$. Then f is a finite-support function

$$f : \mathcal{M}_Y \rightarrow R[X],$$

and for each $\beta \in \mathcal{M}_Y$, $f(\beta)$ is a finite-support function

$$f(\beta) : \mathcal{M}_X \rightarrow R.$$

Define $\Phi(f) : \mathcal{M}_{X \sqcup Y} \rightarrow R$ by

$$\Phi(f)(\gamma) := f(\beta)(\alpha),$$

where $(\alpha, \beta) = \Psi^{-1}(\gamma)$.

One checks that $\Phi(f)$ has finite support, hence $\Phi(f) \in R[X \sqcup Y]$.

Inverse map. Given $g \in R[X \sqcup Y]$, define $\tilde{\Phi}(g) : \mathcal{M}_Y \rightarrow R[X]$ by

$$\tilde{\Phi}(g)(\beta)(\alpha) := g(\Psi(\alpha, \beta)).$$

One checks that this has finite support and defines an element of $R[X][Y]$. It is immediate that Φ and $\tilde{\Phi}$ are inverse bijections.

Additivity. For $f_1, f_2 \in R[X][Y]$ and $\gamma \in \mathcal{M}_{X \sqcup Y}$, write $\Psi^{-1}(\gamma) = (\alpha, \beta)$. Then

$$\Phi(f_1 + f_2)(\gamma) = (f_1 + f_2)(\beta)(\alpha) = f_1(\beta)(\alpha) + f_2(\beta)(\alpha) = \Phi(f_1)(\gamma) + \Phi(f_2)(\gamma).$$

Multiplicativity. Fix $f_1, f_2 \in R[X][Y]$ and $\gamma \in \mathcal{M}_{X \sqcup Y}$, and write

$$\Psi^{-1}(\gamma) = (\alpha, \beta).$$

We compute

$$\Phi(f_1 f_2)(\gamma) = (f_1 f_2)(\beta)(\alpha).$$

By the multiplication rule in $R[X][Y]$,

$$(f_1 f_2)(\beta) = \sum_{\beta_1 + \beta_2 = \beta} f_1(\beta_1) f_2(\beta_2),$$

so

$$\Phi(f_1 f_2)(\gamma) = \sum_{\beta_1 + \beta_2 = \beta} (f_1(\beta_1) f_2(\beta_2))(\alpha).$$

Now applying multiplication in $R[X]$, we get

$$(f_1(\beta_1) f_2(\beta_2))(\alpha) = \sum_{\alpha_1 + \alpha_2 = \alpha} f_1(\beta_1)(\alpha_1) f_2(\beta_2)(\alpha_2).$$

Hence

$$\Phi(f_1 f_2)(\gamma) = \sum_{\beta_1 + \beta_2 = \beta} \sum_{\alpha_1 + \alpha_2 = \alpha} f_1(\beta_1)(\alpha_1) f_2(\beta_2)(\alpha_2).$$

Reindexing. Define

$$A := \{(\alpha_1, \alpha_2, \beta_1, \beta_2) : \alpha_1 + \alpha_2 = \alpha, \beta_1 + \beta_2 = \beta\},$$

and

$$B := \{(\gamma_1, \gamma_2) : \gamma_1 + \gamma_2 = \gamma\}.$$

Define

$$T : A \rightarrow B$$

by

$$T(\alpha_1, \alpha_2, \beta_1, \beta_2) = (\Psi(\alpha_1, \beta_1), \Psi(\alpha_2, \beta_2)).$$

Then T is a bijection, with inverse

$$T^{-1}(\gamma_1, \gamma_2) = (\gamma_1|_X, \gamma_2|_X, \gamma_1|_Y, \gamma_2|_Y).$$

Compatibility of summands. For $(\alpha_1, \alpha_2, \beta_1, \beta_2) \in A$, let

$$\gamma_1 := \Psi(\alpha_1, \beta_1), \quad \gamma_2 := \Psi(\alpha_2, \beta_2).$$

Then

$$\Phi(f_1)(\gamma_1) = f_1(\beta_1)(\alpha_1), \quad \Phi(f_2)(\gamma_2) = f_2(\beta_2)(\alpha_2),$$

and hence

$$f_1(\beta_1)(\alpha_1) f_2(\beta_2)(\alpha_2) = \Phi(f_1)(\gamma_1) \Phi(f_2)(\gamma_2).$$

Thus the summand is preserved under T .

Since T is a bijection and preserves summands, we obtain

$$\sum_{\substack{\alpha_1 + \alpha_2 = \alpha \\ \beta_1 + \beta_2 = \beta}} f_1(\beta_1)(\alpha_1) f_2(\beta_2)(\alpha_2) = \sum_{\gamma_1 + \gamma_2 = \gamma} \Phi(f_1)(\gamma_1) \Phi(f_2)(\gamma_2).$$

Therefore

$$\Phi(f_1 f_2)(\gamma) = (\Phi(f_1) \Phi(f_2))(\gamma).$$

Hence Φ is a ring isomorphism. □

2 Construction: Polynomial Ring in a set of finite variables, defined recursively

Lemma 2.1. (Recursive construction of iterated polynomial ambient sets via Replacement)

Let R be a set. Then there exists a function f such that

$$\text{dom}(f) = \omega, \quad f(0) = R, \quad \forall n \in \omega, \quad f(n+1) = f(n)^\omega.$$

Moreover, there exists a set A such that $f(n) \subseteq A$ for all $n \in \omega$.

Proof. **Step 1: Define the recursion formula.**

Define $\varphi(n, y)$ to be the formula:

$$\exists g \left(\text{Func}(g) \wedge \text{dom}(g) = n+1 \wedge g(0) = R \wedge \forall k \in n \left(g(k+1) = g(k)^\omega \right) \wedge y = g(n) \right).$$

Step 2: Verify uniqueness.

We claim:

$$\forall n \in \omega \quad \forall y_1, y_2 \quad (\varphi(n, y_1) \wedge \varphi(n, y_2) \Rightarrow y_1 = y_2).$$

Indeed, if g_1, g_2 are two functions witnessing $\varphi(n, y_1)$ and $\varphi(n, y_2)$, then by induction on $k \leq n$, one shows $g_1(k) = g_2(k)$. In particular $y_1 = g_1(n) = g_2(n) = y_2$.

Step 3: Verify existence.

We prove:

$$\forall n \in \omega \quad \exists y \quad \varphi(n, y)$$

by induction on n .

Base case: For $n = 0$, define

$$g_0 := \{(0, R)\}.$$

Then g_0 satisfies the defining conditions, so $\varphi(0, R)$ holds.

Inductive step: Assume for n there exists g with domain $n+1$ satisfying the recursion. Define

$$h := g \cup \{(n+1, g(n)^\omega)\}.$$

Then h is a function with domain $n+2$, and satisfies the recursion at all stages $k \leq n$. Thus $\varphi(n+1, h(n+1))$ holds.

Step 4: Apply Replacement.

By Step 2, $\varphi(n, y)$ is functional in y . By the Axiom of Replacement, the set

$$B := \{y \mid \exists n \in \omega \quad \varphi(n, y)\}$$

exists.

Step 5: Construct the function.

Define

$$f := \{(n, y) \in \omega \times B \mid \varphi(n, y)\}.$$

By Separation, f is a set. By uniqueness (Step 2), f is a function. By existence (Step 3), $\text{dom}(f) = \omega$.

We show that the function f satisfies the recursive property

$$f(0) = R \quad \text{and} \quad \forall n \in \omega, \quad f(n+1) = f(n)^\omega.$$

First, we prove $f(0) = R$. Define $g := \{(0, R)\}$. Then g is a function with

$$\text{dom}(g) = 1 = 0+1, \quad g(0) = R,$$

and the condition

$$\forall k \in 0 \quad [g(k+1) = g(k)^\omega]$$

holds vacuously. Hence $\varphi(0, R)$ holds. By definition of f , we also have $\varphi(0, f(0))$. By uniqueness (Step 2), it follows that

$$f(0) = R.$$

Next, define

$$T := \{n \in \omega : f(n+1) = f(n)^\omega\}.$$

We prove $T = \omega$ by induction.

Base case. We show $0 \in T$, i.e., $f(1) = f(0)^\omega$. Since $\varphi(1, f(1))$, there exists a function g such that

$$\text{dom}(g) = 2, \quad g(0) = R, \quad \forall k \in 1 [g(k+1) = g(k)^\omega], \quad f(1) = g(1).$$

Since $0 \in 1$, we have $g(1) = g(0)^\omega = R^\omega$. Thus

$$f(1) = R^\omega = f(0)^\omega,$$

so $0 \in T$.

Inductive step. Suppose $n \in T$. We show $n+1 \in T$, i.e., $f(n+2) = f(n+1)^\omega$. Since $\varphi(n+2, f(n+2))$, there exists a function g such that

$$\text{dom}(g) = n+3, \quad g(0) = R, \quad \forall k \in n+2 [g(k+1) = g(k)^\omega], \quad f(n+2) = g(n+2).$$

Since $n+1 \in n+2$, we have

$$f(n+2) = g(n+2) = g(n+1)^\omega.$$

Consider the restriction $g \upharpoonright (n+2)$. Then $g \upharpoonright (n+2)$ **witnesses** $\varphi(n+1, g(n+1))$. By definition of f , we also have $\varphi(n+1, f(n+1))$. By uniqueness (Step 2),

$$g(n+1) = f(n+1).$$

Hence

$$f(n+2) = f(n+1)^\omega,$$

so $n+1 \in T$.

By induction, $T = \omega$. Therefore

$$\forall n \in \omega, \quad f(n+1) = f(n)^\omega.$$

Step 6: Construct the ambient set.

Define

$$A := \bigcup B.$$

Then A is a set by Union, and for every $n \in \omega$,

$$f(n) \in B \Rightarrow f(n) \subseteq A.$$

This completes the proof. □

Construction: Polynomial Ring in a finite set of variables

Let $var : \omega - \{0_\omega\} \rightarrow Var$ be an injective map, R be a ring.

Apply previous lemma, we obtain set A and map $f : \omega \rightarrow B$.

Let $V := \mathcal{P}(A)$. We define $F : V \rightarrow V$ as,

$$S \mapsto \begin{cases} \{g \in S^\omega \mid \text{supp}(g) \text{ finite}\}, & \text{if } \exists (n \in \omega) S \subseteq f(n) \\ \emptyset, & \text{otherwise} \end{cases}$$

Apply recursion on ω and obtain map $h : \omega \rightarrow V$, s.t.

$$h(0) = R, \quad h(n^+) = F(h(n))$$

We can prove by induction on ω , that $\forall k \in \omega, h(k) \subseteq f(n)$, for some $n \in \omega$.

Examples: $h(0) = R, h(1) = F(R) = R[x_1], h(2) = F(R[x_1]) = R[x_1][x_2]$, etc.

Construction (continued): with binary operations

Define

$$V' := \mathcal{P}(A) \times \mathcal{P}(A^2 \times A) \times \mathcal{P}(A^2 \times A).$$

For $(S, \alpha, \mu) \in V'$, define

$$F'(S, \alpha, \mu) = \begin{cases} (F(S), +_{F(S)}, *_{F(S)}), & \text{if } (S, \alpha, \mu) \text{ codes a ring,} \\ (\emptyset, \emptyset, \emptyset), & \text{otherwise,} \end{cases}$$

where $+_{F(S)}, *_{F(S)}$ are the usual polynomial addition and multiplication induced from α and μ (as in construction of $R[x]$ of one variable).

By Lemma 1.1, there exists a function $h' : \omega \rightarrow V'$ such that

$$h'(0) = (R, +_R, *_R)$$

and

$$\forall n \in \omega, \quad h'(n+1) = F'(h'(n)).$$

We now show by induction on ω that for each $n \in \omega$, if

$$h'(n) = (S, +_S, *_S),$$

then $h'(n)$ codes a ring and

$$h'(n+1) = (F(S), +_{F(S)}, *_{F(S)}),$$

i.e. $h'(n+1)$ is the code of the polynomial ring in one variable over the ring coded by $h'(n)$.

Theorem 2.1. (*constructions are isomorphic on finite set of variables*)
Let $n \in \omega$, and $|X| = n$. Then,

$$R[X] \cong h(n)$$

Proof. We argue by induction on n .

Base case: $n = 0$. Then $X = \emptyset$, so by convention

$$R[X] = R[\emptyset] = R.$$

On the other hand, by definition of the recursive construction,

$$h(0) = R.$$

Hence

$$R[X] \cong h(0).$$

Inductive step: Assume that for some $n \in \omega$, whenever Y is a finite set with $|Y| = n$, one has

$$R[Y] \cong h(n).$$

Let X be a finite set with $|X| = n + 1$. Choose $x \in X$, and put

$$Y := X \setminus \{x\}.$$

Then $|Y| = n$, so by the induction hypothesis,

$$R[Y] \cong h(n).$$

Since Y and $\{x\}$ are disjoint and $X = Y \sqcup \{x\}$, Theorem 1.2 yields

$$R[X] = R[Y \sqcup \{x\}] \cong R[Y][\{x\}].$$

Identifying $R[Y][\{x\}]$ with the one-variable polynomial ring over $R[Y]$, we obtain

$$R[X] \cong R[Y][x].$$

Now apply the induction hypothesis:

$$R[Y][x] \cong h(n)[x].$$

By the recursive definition of h ,

$$h(n+1) = F(h(n))$$

and, since $h(n) \subseteq f(n)$, the first clause in the definition of F applies. Therefore

$$h(n+1) = \{g \in h(n)^\omega : \text{supp}(g) \text{ is finite}\} = h(n)[x].$$

Hence

$$R[X] \cong h(n+1).$$

This completes the induction. □

3 Imbedding and Projection

Lemma 3.1. (*Natural embedding into a larger polynomial ring*)

Let R be a ring, and let $Y \subseteq X$. Then there is a natural ring embedding

$$\iota_{Y,X} : R[Y] \hookrightarrow R[X].$$

Proof. Recall the constructions:

$$M_Y := \{\alpha : Y \rightarrow \omega \mid \alpha \text{ has finite support}\}, \quad M_X := \{\beta : X \rightarrow \omega \mid \beta \text{ has finite support}\},$$

$$R[Y] := \{f : M_Y \rightarrow R \mid \text{supp}(f) \text{ is finite}\}, \quad R[X] := \{g : M_X \rightarrow R \mid \text{supp}(g) \text{ is finite}\}.$$

For each $\alpha \in M_Y$, define its zero-extension $\tilde{\alpha} \in M_X$ by

$$\tilde{\alpha}(x) := \begin{cases} \alpha(x), & x \in Y, \\ 0_\omega, & x \in X \setminus Y. \end{cases}$$

We first verify that $\tilde{\alpha} \in M_X$ is well-defined. Observe,

$$\text{supp}(\tilde{\alpha}) = \{x \in X : \tilde{\alpha}(x) \neq 0\} = \{x \in Y : \alpha(x) \neq 0\} = \text{supp}(\alpha) \implies \tilde{\alpha} \in M_X$$

Now define $\iota_{Y,X} : R[Y] \rightarrow R[X]$ as,

$$f \mapsto \left(\beta \mapsto \begin{cases} f(\alpha), & \text{if } \beta = \tilde{\alpha} \text{ for some } \alpha \in M_Y \\ 0_R, & \text{otherwise} \end{cases} \right)$$

Step 1: the definition is unambiguous.

Suppose $\beta = \tilde{\alpha}_1 = \tilde{\alpha}_2$ for some $\alpha_1, \alpha_2 \in M_Y$. Then for every $y \in Y$,

$$\alpha_1(y) = \tilde{\alpha}_1(y) = \beta(y) = \tilde{\alpha}_2(y) = \alpha_2(y).$$

Hence $\alpha_1 = \alpha_2$. Therefore for each $\beta \in M_X$, there is at most one $\alpha \in M_Y$ such that $\beta = \tilde{\alpha}$. So the above formula for $\iota_{Y,X}(f)(\beta)$ is well-defined.

Step 2: for each $f \in R[Y]$, the function $\iota_{Y,X}(f)$ lies in $R[X]$.

Take $f \in R[Y]$. We must show that $\iota_{Y,X}(f) : M_X \rightarrow R$ has finite support.

Let $\beta \in M_X$ satisfy $\iota_{Y,X}(f)(\beta) \neq 0$. By definition of $\iota_{Y,X}$, this implies that there exists $\alpha \in M_Y$ such that

$$\beta = \tilde{\alpha} \quad \text{and} \quad f(\alpha) \neq 0.$$

Thus $\alpha \in \text{supp}(f)$, and therefore

$$\beta = \tilde{\alpha} \in \{\gamma \in M_X \mid \exists(\alpha \in \text{supp}(f)) \gamma = \tilde{\alpha}\}$$

So we have shown that

$$\text{supp}(\iota_{Y,X}(f)) \subseteq \{\gamma \in M_X \mid \exists(\alpha \in \text{supp}(f)) \gamma = \tilde{\alpha}\}$$

Since $\text{supp}(f)$ is finite, the set on the right is finite. Hence $\text{supp}(\iota_{Y,X}(f))$ is finite.

Therefore, $\iota_{Y,X}$ is well-defined as a map $R[Y] \rightarrow R[X]$.

Step 3: $\iota_{Y,X}$ preserves addition.

Let $f, g \in R[Y]$. We must show

$$\iota_{Y,X}(f + g) = \iota_{Y,X}(f) + \iota_{Y,X}(g).$$

Fix $\beta \in M_X$. We consider two cases.

Case 1: $\beta = \tilde{\alpha}$ for some $\alpha \in M_Y$.

$$\implies \iota_{Y,X}(f + g)(\beta) = (f + g)(\alpha) = f(\alpha) + g(\alpha) = \iota_{Y,X}(f)(\beta) + \iota_{Y,X}(g)(\beta).$$

Case 2: $\beta \neq \tilde{\alpha}$ for every $\alpha \in M_Y$.

$$\implies \iota_{Y,X}(f+g)(\beta) = 0, \quad \iota_{Y,X}(f)(\beta) = 0, \quad \iota_{Y,X}(g)(\beta) = 0.$$

Hence again

$$\iota_{Y,X}(f+g)(\beta) = \iota_{Y,X}(f)(\beta) + \iota_{Y,X}(g)(\beta).$$

Since the equality holds for every $\beta \in M_X$, we conclude

$$\iota_{Y,X}(f+g) = \iota_{Y,X}(f) + \iota_{Y,X}(g).$$

Step 4: $\iota_{Y,X}$ preserves multiplication.

Let $f, g \in R[Y]$. We must show

$$\iota_{Y,X}(fg) = \iota_{Y,X}(f) \iota_{Y,X}(g).$$

Fix $\beta \in M_X$.

Again we split into two cases.

Case 1: $\beta = \tilde{\alpha}$ for some $\alpha \in M_Y$.

Then, by the convolution rule for multiplication in $R[X]$,

$$(\iota_{Y,X}(f)\iota_{Y,X}(g))(\tilde{\alpha}) = \sum_{\beta_1 + \beta_2 = \tilde{\alpha}} \iota_{Y,X}(f)(\beta_1) \iota_{Y,X}(g)(\beta_2)$$

We now analyze which pairs $(\beta_1, \beta_2) \in M_X^2$ can contribute nonzero terms. Suppose,

$$\beta_1 + \beta_2 = \tilde{\alpha} \quad \text{and} \quad \iota_{Y,X}(f)(\beta_1) \neq 0, \quad \iota_{Y,X}(g)(\beta_2) \neq 0$$

Then, by definition of $\iota_{Y,X}$, there exist $\alpha_1, \alpha_2 \in M_Y$ such that $\beta_1 = \widetilde{\alpha_1}$, $\beta_2 = \widetilde{\alpha_2}$. Hence,

$$\widetilde{\alpha_1} + \widetilde{\alpha_2} = \tilde{\alpha}.$$

Evaluating both sides at each $y \in Y$ gives

$$\alpha_1(y) + \alpha_2(y) = \alpha(y) \implies \alpha_1 + \alpha_2 = \alpha$$

We construct

1. $\mathcal{B} := \{(\beta_1, \beta_2) \in M_X^2 \mid [\beta_1 + \beta_2 = \tilde{\alpha}] \wedge [\iota_{Y,X}(f)(\beta_1), \iota_{Y,X}(g)(\beta_2) \neq 0]\}$
2. $\mathcal{A} := \{(\widetilde{\alpha_1}, \widetilde{\alpha_2}) \in M_X^2 \mid \alpha_1, \alpha_2 \in M_Y \wedge \alpha_1 + \alpha_2 = \alpha \wedge f(\alpha_1), g(\alpha_2) \neq 0\}$

Observe, we have $\mathcal{B} = \mathcal{A}$. Then,

$$(\iota_{Y,X}(f)\iota_{Y,X}(g))(\tilde{\alpha}) = \sum_{(\beta_1, \beta_2) \in \mathcal{B}} \iota_{Y,X}(f)(\beta_1) \iota_{Y,X}(g)(\beta_2) \tag{1}$$

$$= \sum_{(\alpha_1, \alpha_2) \in \mathcal{A}} \iota_{Y,X}(f)(\widetilde{\alpha_1}) \iota_{Y,X}(g)(\widetilde{\alpha_2}) \tag{2}$$

$$= \sum_{\alpha_1 + \alpha_2 = \alpha} f(\alpha_1)g(\alpha_2) \tag{3}$$

$$= (fg)(\alpha) \tag{4}$$

$$= \iota_{Y,X}(fg)(\tilde{\alpha}) \tag{5}$$

Case 2: $\beta \neq \tilde{\alpha}$ for every $\alpha \in M_Y$.

We claim

$$(\iota_{Y,X}(f)\iota_{Y,X}(g))(\beta) = 0.$$

Indeed, suppose a term in the convolution sum were nonzero. Then there would exist $\beta_1, \beta_2 \in M_X$ such that

$$\beta_1 + \beta_2 = \beta, \quad \iota_{Y,X}(f)(\beta_1) \neq 0, \quad \iota_{Y,X}(g)(\beta_2) \neq 0.$$

By definition of $\iota_{Y,X}$, this implies

$$\beta_1 = \widetilde{\alpha_1}, \quad \beta_2 = \widetilde{\alpha_2}$$

for some $\alpha_1, \alpha_2 \in M_Y$. Therefore

$$\beta = \beta_1 + \beta_2 = \widetilde{\alpha_1} + \widetilde{\alpha_2} = \widetilde{\alpha_1 + \alpha_2},$$

contrary to the assumption that β is not of the form $\widetilde{\alpha}$. So every summand is zero, and hence

$$(\iota_{Y,X}(f)\iota_{Y,X}(g))(\beta) = 0.$$

On the other hand, by definition,

$$\iota_{Y,X}(fg)(\beta) = 0$$

as well, since β is not an extended exponent. Thus

$$\iota_{Y,X}(fg)(\beta) = (\iota_{Y,X}(f)\iota_{Y,X}(g))(\beta).$$

Since the equality holds for every $\beta \in M_X$, we conclude

$$\iota_{Y,X}(fg) = \iota_{Y,X}(f)\iota_{Y,X}(g).$$

Step 5: $\iota_{Y,X}$ preserves the multiplicative identity (if R has 1)

Let $1_{R[Y]}$ and $1_{R[X]}$ denote the identity polynomials in $R[Y]$ and $R[X]$, respectively. These are the functions supported only at the zero exponent functions $0_Y \in M_Y$ and $0_X \in M_X$.

Since $\widetilde{0_Y} = 0_X$, we have

$$\iota_{Y,X}(1_{R[Y]})(0_X) = 1_{R[Y]}(0_Y) = 1_R.$$

If $\beta \neq 0_X$, then either $\beta = \widetilde{\alpha}$ for some $\alpha \neq 0_Y$, in which case

$$\iota_{Y,X}(1_{R[Y]})(\beta) = 1_{R[Y]}(\alpha) = 0,$$

or else β is not of the form $\widetilde{\alpha}$, in which case

$$\iota_{Y,X}(1_{R[Y]})(\beta) = 0$$

by definition.

Therefore $\iota_{Y,X}(1_{R[Y]})$ is supported only at 0_X and takes value 1_R there. Hence

$$\iota_{Y,X}(1_{R[Y]}) = 1_{R[X]}.$$

So $\iota_{Y,X}$ is a ring homomorphism.

Step 6: $\iota_{Y,X}$ is injective.

Suppose $\iota_{Y,X}(f) = \iota_{Y,X}(g)$, for some $f, g \in R[Y]$. We must prove $f = g$.

Let $\alpha \in M_Y$ be arbitrary. Then

$$f(\alpha) = \iota_{Y,X}(f)(\widetilde{\alpha}) = \iota_{Y,X}(g)(\widetilde{\alpha}) = g(\alpha).$$

Since this holds for every $\alpha \in M_Y$, we obtain

$$f = g.$$

Thus $\iota_{Y,X}$ is injective.

Hence $\iota_{Y,X} : R[Y] \rightarrow R[X]$ is an injective ring homomorphism, i.e. a ring embedding. □

Lemma 3.2. (Projection of polynomial rings)

Let R be a ring, and let $Y \subseteq X$. Then there exists a ring homomorphism

$$\pi_{X,Y} : R[X] \rightarrow R[Y], \quad \text{s.t. it satisfies}$$

1. If R has 1, then $\pi_{X,Y}$ preserves multiplicative identity
2. $\pi_{X,Y} \circ \iota_{Y,X} = \text{id}_{R[Y]}$

Proof. We define $\pi_{X,Y} : R[X] \rightarrow R[Y]$ as,

$$f \mapsto (\alpha \mapsto f(\tilde{\alpha}))$$

where $\tilde{\alpha} \in M_X$ denotes the extension of α from Y to X by zero outside Y .

Step 1: $\pi_{X,Y}$ is well-defined.

Fix $f \in R[X]$. We show that $\pi_{X,Y}(f)$ has finite support. By definition,

$$\text{supp}(\pi_{X,Y}(f)) = \{ \alpha \in M_Y \mid \pi_{X,Y}(f)(\alpha) \neq 0 \} = \{ \alpha \in M_Y \mid f(\tilde{\alpha}) \neq 0 \}.$$

Consider the map

$$\Phi : \text{supp}(\pi_{X,Y}(f)) \rightarrow \text{supp}(f), \quad \alpha \mapsto \tilde{\alpha}.$$

This map is injective, because if $\tilde{\alpha}_1 = \tilde{\alpha}_2$, then evaluating both sides on each $y \in Y$ gives

$$\alpha_1(y) = \alpha_2(y),$$

so $\alpha_1 = \alpha_2$. Since $\text{supp}(f)$ is finite, it follows that $\text{supp}(\pi_{X,Y}(f))$ is finite. Hence $\pi_{X,Y}(f) \in R[Y]$. Therefore $\pi_{X,Y}$ is well-defined.

Step 2: $\pi_{X,Y}$ preserves addition.

Let $f, g \in R[X]$, and let $\alpha \in M_Y$. Then

$$\pi_{X,Y}(f + g)(\alpha) = (f + g)(\tilde{\alpha}) = f(\tilde{\alpha}) + g(\tilde{\alpha}) = \pi_{X,Y}(f)(\alpha) + \pi_{X,Y}(g)(\alpha).$$

Since this holds for every $\alpha \in M_Y$, we conclude

$$\pi_{X,Y}(f + g) = \pi_{X,Y}(f) + \pi_{X,Y}(g).$$

Step 3: $\pi_{X,Y}$ preserves multiplication.

Let $f, g \in R[X]$. We show that

$$\pi_{X,Y}(fg) = \pi_{X,Y}(f)\pi_{X,Y}(g).$$

Fix $\alpha \in M_Y$. Then by the convolution rule for multiplication in $R[X]$,

$$\pi_{X,Y}(fg)(\alpha) = (fg)(\tilde{\alpha}) = \sum_{\beta_1 + \beta_2 = \tilde{\alpha}} f(\beta_1)g(\beta_2).$$

We now determine which pairs $(\beta_1, \beta_2) \in M_X^2$ can occur in this sum.

Suppose $\beta_1, \beta_2 \in M_X$ satisfy

$$\beta_1 + \beta_2 = \tilde{\alpha}.$$

Let $x \in X \setminus Y$. Since $\tilde{\alpha}(x) = 0$, we have

$$\beta_1(x) + \beta_2(x) = 0.$$

But $\beta_1(x), \beta_2(x) \in \omega$, so necessarily

$$\beta_1(x) = 0 \quad \text{and} \quad \beta_2(x) = 0.$$

Since this holds for every $x \in X \setminus Y$, both β_1 and β_2 vanish outside Y . Hence there exist unique $\alpha_1, \alpha_2 \in M_Y$ such that

$$\beta_1 = \tilde{\alpha}_1, \quad \beta_2 = \tilde{\alpha}_2.$$

Substituting into $\beta_1 + \beta_2 = \tilde{\alpha}$ gives

$$\tilde{\alpha}_1 + \tilde{\alpha}_2 = \tilde{\alpha}.$$

Evaluating both sides at each $y \in Y$ yields

$$\alpha_1(y) + \alpha_2(y) = \alpha(y),$$

so

$$\alpha_1 + \alpha_2 = \alpha.$$

Conversely, if $\alpha_1, \alpha_2 \in M_Y$ satisfy $\alpha_1 + \alpha_2 = \alpha$, then

$$\widetilde{\alpha_1} + \widetilde{\alpha_2} = \widetilde{\alpha}.$$

Therefore the pairs $(\beta_1, \beta_2) \in M_X^2$ satisfying

$$\beta_1 + \beta_2 = \widetilde{\alpha}$$

are exactly the pairs of the form

$$(\widetilde{\alpha_1}, \widetilde{\alpha_2}) \quad \text{where} \quad \alpha_1, \alpha_2 \in M_Y, \quad \alpha_1 + \alpha_2 = \alpha.$$

Hence

$$\pi_{X,Y}(fg)(\alpha) = \sum_{\alpha_1 + \alpha_2 = \alpha} f(\widetilde{\alpha_1}) g(\widetilde{\alpha_2}).$$

By definition of $\pi_{X,Y}$, this becomes

$$\pi_{X,Y}(fg)(\alpha) = \sum_{\alpha_1 + \alpha_2 = \alpha} \pi_{X,Y}(f)(\alpha_1) \pi_{X,Y}(g)(\alpha_2).$$

But this is precisely the convolution formula for multiplication in $R[Y]$, so

$$\pi_{X,Y}(fg)(\alpha) = (\pi_{X,Y}(f)\pi_{X,Y}(g))(\alpha).$$

Since this holds for every $\alpha \in M_Y$, we conclude

$$\pi_{X,Y}(fg) = \pi_{X,Y}(f)\pi_{X,Y}(g).$$

Thus $\pi_{X,Y}$ is a ring homomorphism.

Step 4: If R has identity, then $\pi_{X,Y}$ preserves multiplicative identity.

Assume that R has multiplicative identity 1_R . Let $1_{R[X]}$ and $1_{R[Y]}$ denote the identity polynomials in $R[X]$ and $R[Y]$, respectively. These are supported only at the zero exponent functions $0_X \in M_X$ and $0_Y \in M_Y$, and satisfy

$$1_{R[X]}(0_X) = 1_R, \quad 1_{R[Y]}(0_Y) = 1_R.$$

We show that

$$\pi_{X,Y}(1_{R[X]}) = 1_{R[Y]}.$$

Let $\alpha \in M_Y$.

If $\alpha = 0_Y$, then $\widetilde{\alpha} = 0_X$, so

$$\pi_{X,Y}(1_{R[X]})(0_Y) = 1_{R[X]}(0_X) = 1_R.$$

If $\alpha \neq 0_Y$, then $\widetilde{\alpha} \neq 0_X$, so

$$\pi_{X,Y}(1_{R[X]})(\alpha) = 1_{R[X]}(\widetilde{\alpha}) = 0.$$

Thus $\pi_{X,Y}(1_{R[X]})$ is supported only at 0_Y and takes value 1_R there. Therefore

$$\pi_{X,Y}(1_{R[X]}) = 1_{R[Y]}.$$

Step 5: $\pi_{X,Y} \circ \iota_{Y,X} = \text{id}_{R[Y]}$.

Let $h \in R[Y]$. We show that

$$\pi_{X,Y}(\iota_{Y,X}(h)) = h.$$

Fix $\alpha \in M_Y$. Then, by definition of $\pi_{X,Y}$ and $\iota_{Y,X}$,

$$(\pi_{X,Y}(\iota_{Y,X}(h)))(\alpha) = \iota_{Y,X}(h)(\widetilde{\alpha}) = h(\alpha).$$

Since this holds for every $\alpha \in M_Y$, we obtain

$$\pi_{X,Y}(\iota_{Y,X}(h)) = h.$$

Hence

$$\pi_{X,Y} \circ \iota_{Y,X} = \text{id}_{R[Y]}.$$

This completes the proof. □

4 Universal Mapping Property

Theorem 4.1. (*Universal mapping property for polynomial ring in a set of variables*)

Let R, A be rings with identity, let X be a set, let

$$\varphi : R \rightarrow A$$

be a unital ring homomorphism, and let

$$\eta : X \rightarrow A$$

be a function satisfying the following two conditions:

1. for all $\lambda, \mu \in X$,

$$\eta(\lambda)\eta(\mu) = \eta(\mu)\eta(\lambda),$$

2. for all $\lambda \in X$ and all $r \in R$,

$$\eta(\lambda)\varphi(r) = \varphi(r)\eta(\lambda),$$

Then there exists a **unique unital** ring homomorphism

$$\Phi : R[X] \rightarrow A$$

such that

1. $\Phi(r) = \varphi(r)$ for every $r \in R$, where r is viewed as a constant polynomial in $R[X]$;
2. $\Phi(X_\lambda) = \eta(\lambda)$ for every $\lambda \in X$.

Proof. We prove uniqueness first and existence second.

(uniqueness) Let

$$f \in R[X].$$

By construction, f is a finitely supported function on the set M of monomials. Therefore f may be written uniquely as a finite sum

$$f = \sum_{\alpha \in \text{supp}(f)} f(\alpha) X^\alpha,$$

where $\text{supp}(f)$ is finite, $f(\alpha) \in R$, and X^α denotes the monomial corresponding to the exponent function α . Suppose

$$\Phi : R[X] \rightarrow A$$

is a unital ring homomorphism such that

1. $\Phi(r) = \varphi(r)$ for every $r \in R$;
2. $\Phi(X_\lambda) = \eta(\lambda)$ for every $\lambda \in X$.

Take any monomial X^α . Since Φ preserves multiplication and each X_λ is sent to $\eta(\lambda)$, we must have

$$\Phi(X^\alpha) = \prod_{\lambda \in \text{supp}(\alpha)} \eta(\lambda)^{\alpha(\lambda)}.$$

This product is finite because α has finite support. It is unambiguous because the $\eta(\lambda)$'s commute with one another.

Now apply Φ to the finite sum expression for f . Since Φ preserves addition and sends coefficients by φ , we must have

$$\Phi(f) = \sum_{\alpha \in \text{supp}(f)} \varphi(f(\alpha)) \prod_{\lambda \in \text{supp}(\alpha)} \eta(\lambda)^{\alpha(\lambda)}.$$

So the value of $\Phi(f)$ is forced for every polynomial f . Therefore there is at most one homomorphism with the required properties.

(existence) We now define a map

$$\Phi : R[X] \rightarrow A$$

by exactly the formula forced in the uniqueness step.

Let $f \in R[X]$. Define

$$\Phi(f) := \sum_{\alpha \in \text{supp}(f)} \varphi(f(\alpha)) \prod_{\lambda \in \text{supp}(\alpha)} \eta(\lambda)^{\alpha(\lambda)}.$$

We check that this is well-defined.

1. Since f has finite support, the outer sum is finite.
2. Since each α has finite support, the inner product is finite.
3. Since A has identity, the product for the zero exponent function is interpreted in the usual way as 1_A .
4. Since the $\eta(\lambda)$'s commute pairwise, the product depends only on α , not on any ordering of its support.
5. Since each $\eta(\lambda)$ commutes with $\varphi(R)$, the coefficient term $\varphi(f(\alpha))$ may be placed next to the monomial product without ambiguity.

(additivity) Let $f, g \in R[X]$. Since addition in $R[X]$ is defined pointwise on coefficient functions,

$$(f + g)(\alpha) = f(\alpha) + g(\alpha)$$

for every monomial α .

Therefore

$$\Phi(f + g) = \sum_{\alpha} \varphi(f(\alpha) + g(\alpha)) \prod_{\lambda} \eta(\lambda)^{\alpha(\lambda)}.$$

Since φ is additive,

$$\varphi(f(\alpha) + g(\alpha)) = \varphi(f(\alpha)) + \varphi(g(\alpha)).$$

Because the sum is finite, this splits into two finite sums:

$$\Phi(f + g) = \Phi(f) + \Phi(g).$$

(multiplicativity) Let $f, g \in R[X]$. We must show

$$\Phi(fg) = \Phi(f)\Phi(g).$$

By the multiplication rule in the construction,

$$(fg)(\gamma) = \sum_{\alpha + \beta = \gamma} f(\alpha)g(\beta),$$

where the sum is finite because f and g have finite support.

Therefore

$$\Phi(fg) = \sum_{\gamma} \varphi((fg)(\gamma)) \prod_{\lambda \in \text{supp}(\gamma)} \eta(\lambda)^{\gamma(\lambda)}.$$

Substituting the formula for $(fg)(\gamma)$, we get

$$\Phi(fg) = \sum_{\gamma} \varphi \left(\sum_{\alpha + \beta = \gamma} f(\alpha)g(\beta) \right) \prod_{\lambda} \eta(\lambda)^{\gamma(\lambda)}.$$

Since φ is additive,

$$\Phi(fg) = \sum_{\gamma} \sum_{\alpha + \beta = \gamma} \varphi(f(\alpha)g(\beta)) \prod_{\lambda} \eta(\lambda)^{\gamma(\lambda)}.$$

Since φ is multiplicative,

$$\varphi(f(\alpha)g(\beta)) = \varphi(f(\alpha))\varphi(g(\beta)).$$

Now use the commutation assumptions. Because the $\eta(\lambda)$'s commute with one another, we have

$$\prod_{\lambda} \eta(\lambda)^{\alpha(\lambda)+\beta(\lambda)} = \left(\prod_{\lambda} \eta(\lambda)^{\alpha(\lambda)} \right) \left(\prod_{\lambda} \eta(\lambda)^{\beta(\lambda)} \right).$$

Because each $\eta(\lambda)$ commutes with $\varphi(R)$, we may move coefficient images past these products freely. Therefore each summand becomes

$$\varphi(f(\alpha)) \left(\prod_{\lambda} \eta(\lambda)^{\alpha(\lambda)} \right) \varphi(g(\beta)) \left(\prod_{\lambda} \eta(\lambda)^{\beta(\lambda)} \right).$$

Thus

$$\Phi(fg) = \sum_{\alpha, \beta} \left[\varphi(f(\alpha)) \prod_{\lambda} \eta(\lambda)^{\alpha(\lambda)} \right] \left[\varphi(g(\beta)) \prod_{\lambda} \eta(\lambda)^{\beta(\lambda)} \right].$$

Since all sums are finite, this is exactly the product of the two finite sums defining $\Phi(f)$ and $\Phi(g)$. Hence

$$\Phi(fg) = \Phi(f)\Phi(g).$$

(preservation of coefficients) Let $r \in R$, viewed as a constant polynomial. Then its coefficient function has value r at the zero exponent function and value 0 everywhere else.

Therefore in the defining formula for $\Phi(r)$, there is only one nonzero summand, and the corresponding monomial product is the empty product, which is 1_A . So

$$\Phi(r) = \varphi(r) \cdot 1_A = \varphi(r).$$

(values on the variables) Fix $\lambda \in X$. The variable polynomial X_{λ} is the polynomial whose coefficient function takes the value 1_R at the exponent function δ_{λ} defined by

$$\delta_{\lambda}(\lambda) = 1, \quad \delta_{\lambda}(\mu) = 0 \text{ for } \mu \neq \lambda,$$

and takes the value 0_R at all other exponents.

Therefore the defining sum for $\Phi(X_{\lambda})$ has only one nonzero term, namely

$$\Phi(X_{\lambda}) = \varphi(1_R)\eta(\lambda).$$

Since φ is unital,

$$\varphi(1_R) = 1_A.$$

Hence

$$\Phi(X_{\lambda}) = 1_A\eta(\lambda) = \eta(\lambda).$$

(preservation of identity) Finally, the multiplicative identity $1_{R[X]}$ is the constant polynomial 1_R . Therefore

$$\Phi(1_{R[X]}) = \Phi(1_R) = \varphi(1_R) = 1_A.$$

So Φ is a unital ring homomorphism.

We have constructed a unital ring homomorphism

$$\Phi : R[X] \rightarrow A$$

such that

1. $\Phi(r) = \varphi(r)$ for all $r \in R$;
2. $\Phi(X_{\lambda}) = \eta(\lambda)$ for all $\lambda \in X$.

Uniqueness was proved at the beginning. Therefore such a homomorphism exists and is unique. $\square$

5 Union of Polynomial Rings of a finite subset of the variables

Theorem 5.1. (*union of finite subset of variables*)

Let R be a ring, and let X be a set. Then

$$R[X] = \bigcup_{Y \subseteq X, Y \text{ finite}} \iota_{Y,X}(R[Y])$$

Proof. We distinguish two cases.

Case 1. $X = \emptyset$.

By the edge-case convention in the construction,

$$R[\emptyset] = R.$$

The only finite subset of $\emptyset$ is $\emptyset$ itself, so

$$\bigcup_{Y \subseteq \emptyset, Y \text{ finite}} \iota_{Y,\emptyset}(R[Y]) = \iota_{\emptyset,\emptyset}(R[\emptyset]) = R[\emptyset].$$

Case 2. $X \neq \emptyset$.

Let $f \in R[X]$. We show that

$$f \in \iota_{Y,X}(R[Y])$$

for some finite subset $Y \subseteq X$.

By construction, f has finite support as a function on the monomial set M_X . For each monomial $\alpha \in \text{supp}(f)$, the exponent function $\alpha : X \rightarrow \omega$ has finite support in X . Define

$$Y_0 := \bigcup_{\alpha \in \text{supp}(f)} \text{supp}(\alpha).$$

Since $\text{supp}(f)$ is finite and each $\text{supp}(\alpha)$ is finite, the set Y_0 is finite.

Now every monomial α occurring in f vanishes outside Y_0 . Therefore f depends only on variables from Y_0 . Equivalently, there exists a polynomial $g \in R[Y_0]$ whose extension by zero to X is exactly f . Thus

$$f = \iota_{Y_0,X}(g),$$

so

$$f \in \iota_{Y_0,X}(R[Y_0]).$$

Since $f \in R[X]$ was arbitrary, we obtain

$$R[X] \subseteq \bigcup_{Y \subseteq X, Y \text{ finite}} \iota_{Y,X}(R[Y]).$$

The reverse inclusion is immediate from the definition of $\iota_{Y,X}$. □

5.1 Leading monomial, exponent, coefficient

Set-up

Let R be a ring, let

$$Z = \{z_1, \dots, z_n\}$$

be a finite, nonempty set of variables, and let $<$ be a monomial order on ω^n .

For each $\alpha = (\alpha_1, \dots, \alpha_n) \in \omega^n$, write

$$z^\alpha := z_1^{\alpha_1} \dots z_n^{\alpha_n}.$$

Thus every polynomial $f \in R[Z]$ may be written uniquely in the form

$$f = \sum_{\alpha \in \omega^n} c_\alpha z^\alpha,$$

where $c_\alpha \in R$ for all α , and $c_\alpha = 0$ for all but finitely many α .

Definition 5.1. Let $f = \sum_{\alpha \in \omega^n} c_\alpha z^\alpha \in R[Z]$. The support of f is

$$\text{supp}(f) := \{\alpha \in \omega^n : c_\alpha \neq 0\}.$$

Definition 5.2. Let $f \in R[Z]$ be nonzero. Since $\text{supp}(f)$ is a nonempty finite subset of ω^n , it has a unique largest element with respect to the monomial order $<$. We call this element the leading exponent of f , and denote it by

$$LE(f).$$

Definition 5.3. Let $f = \sum_{\alpha \in \omega^n} c_\alpha z^\alpha \in R[Z]$ be nonzero. The leading coefficient of f is the coefficient of the monomial $z^{LE(f)}$, and is denoted by

$$LC(f) := c_{LE(f)}.$$

Lemma 5.1. (basic properties of leading exponents and coefficients)

Let R be an integral domain, let

$$Z = \{z_1, \dots, z_n\}$$

be a finite set of variables, and fix a monomial order on ω^n . If $f, g \in R[Z]$ are nonzero, then

$$LE(fg) = LE(f) + LE(g)$$

and

$$LC(fg) = LC(f)LC(g).$$

Proof. Write

$$f = \sum_{\alpha \in \omega^n} c_\alpha z^\alpha, \quad g = \sum_{\beta \in \omega^n} d_\beta z^\beta,$$

where all but finitely many c_α, d_β are zero.

Let

$$\alpha_0 := LE(f), \quad \beta_0 := LE(g).$$

Then

$$LC(f) = c_{\alpha_0} \neq 0, \quad LC(g) = d_{\beta_0} \neq 0,$$

and every exponent α with $c_\alpha \neq 0$ satisfies

$$\alpha \leq \alpha_0,$$

with equality if and only if $\alpha = \alpha_0$. Similarly, every exponent β with $d_\beta \neq 0$ satisfies

$$\beta \leq \beta_0,$$

with equality if and only if $\beta = \beta_0$.

Now write

$$fg = \sum_{\gamma \in \omega^n} e_\gamma z^\gamma,$$

where

$$e_\gamma = \sum_{\alpha+\beta=\gamma} c_\alpha d_\beta.$$

We first show that every exponent γ occurring in fg satisfies

$$\gamma \leq \alpha_0 + \beta_0.$$

Indeed, if $e_\gamma \neq 0$, then there exist $\alpha, \beta \in \omega^n$ such that

$$\alpha + \beta = \gamma, \quad c_\alpha \neq 0, \quad d_\beta \neq 0.$$

Since $\alpha \leq \alpha_0$ and $\beta \leq \beta_0$, and the monomial order is compatible with addition, we obtain

$$\gamma = \alpha + \beta \leq \alpha_0 + \beta_0.$$

Next, we show that the coefficient of $z^{\alpha_0+\beta_0}$ in fg is exactly

$$c_{\alpha_0} d_{\beta_0}.$$

Certainly the pair (α_0, β_0) contributes the term $c_{\alpha_0} d_{\beta_0}$.

Now suppose $\alpha + \beta = \alpha_0 + \beta_0$ with $c_\alpha \neq 0$ and $d_\beta \neq 0$. If $\alpha < \alpha_0$, then by compatibility of the monomial order with addition,

$$\alpha + \beta < \alpha_0 + \beta.$$

Since $\beta \leq \beta_0$, again by compatibility with addition,

$$\alpha_0 + \beta \leq \alpha_0 + \beta_0.$$

Hence

$$\alpha + \beta < \alpha_0 + \beta_0,$$

contrary to assumption. Therefore $\alpha = \alpha_0$. Similarly, $\beta = \beta_0$.

Thus the only pair contributing to the coefficient of $z^{\alpha_0+\beta_0}$ is (α_0, β_0) , and so that coefficient is exactly

$$c_{\alpha_0} d_{\beta_0} = LC(f)LC(g).$$

Since R is an integral domain and both c_{α_0} and d_{β_0} are nonzero, we have

$$c_{\alpha_0} d_{\beta_0} \neq 0.$$

Therefore the exponent $\alpha_0 + \beta_0$ occurs in fg , and no exponent larger than it occurs. Hence

$$LE(fg) = \alpha_0 + \beta_0 = LE(f) + LE(g).$$

Moreover, the coefficient of $z^{LE(fg)} = z^{\alpha_0+\beta_0}$ is

$$LC(fg) = c_{\alpha_0} d_{\beta_0} = LC(f)LC(g).$$

□

Lemma 5.2. *Let R be an integral domain, let X be a nonempty set, let $Y \subseteq X$, and let*

$$S := \iota_{Y,X}(R[Y]) \subseteq R[X].$$

Then S is factorally closed in $R[X]$. That is, if $f, g \in R[X] \setminus \{0\}$ and

$$fg \in S,$$

then

$$f \in S \quad \text{and} \quad g \in S.$$

Proof. Assume

$$f, g \in R[X] \setminus \{0\} \quad \text{and} \quad fg \in S = \iota_{Y,X}(R[Y]).$$

We must prove that both f and g belong to S .

Suppose, for contradiction, that at least one of f or g does not lie in S . Then there exists a variable

$$x \in X \setminus Y$$

which occurs in some monomial of f or g .

Let $Z \subseteq X$ be the finite set of all variables that occur in f or g . Then $x \in Z$, and both f and g lie in the image of $R[Z]$ inside $R[X]$. Since Z is finite, write

$$Z = \{z_1, \dots, z_n\}$$

with

$$z_1 = x.$$

Fix a monomial order on ω^n .

Now regard f and g as nonzero polynomials in the finite-variable ring $R[Z]$. Since $x = z_1$ occurs in f or g , at least one of the first coordinates of $\text{LE}(f)$ and $\text{LE}(g)$ is positive. Hence the first coordinate of

$$\text{LE}(f) + \text{LE}(g)$$

is positive.

By the leading-exponent multiplicative lemma,

$$\text{LE}(fg) = \text{LE}(f) + \text{LE}(g).$$

Therefore the first coordinate of $\text{LE}(fg)$ is positive. Since $z_1 = x$, this means that the leading exponent of fg has positive exponent at the variable x . Hence x occurs in a monomial of fg .

But $x \notin Y$, so this is impossible if

$$fg \in \iota_{Y,X}(R[Y]),$$

because every monomial occurring in an element of $\iota_{Y,X}(R[Y])$ has exponent 0 on every variable in $X \setminus Y$.

This contradiction shows that no variable in $X \setminus Y$ occurs in f or g . Therefore every monomial of f and of g is supported in Y . Equivalently,

$$f \in \iota_{Y,X}(R[Y]) \quad \text{and} \quad g \in \iota_{Y,X}(R[Y]).$$

Thus S is factorally closed in $R[X]$. □

Corollary 5.1.1. *Let R be a unique factorization domain, and let X be any set. Then $R[X]$ is a unique factorization domain.*

Proof. We distinguish two cases.

Case 1. $X = \emptyset$.

By convention,

$$R[\emptyset] = R.$$

Since R is a unique factorization domain, $R[\emptyset]$ is a unique factorization domain.

Case 2. $X \neq \emptyset$.

We must show that every nonzero nonunit element of $R[X]$ factors into irreducibles, and that such factorizations are unique up to order and associates.

Let

$$p \in R[X]$$

be nonzero and not a unit.

By the union lemma, there exists a finite subset $Y \subseteq X$ such that

$$p \in \iota_{Y,X}(R[Y]).$$

Since Y is finite, repeated application of the finite-variable theorem shows that $R[Y]$ is a unique factorization domain. Since $\iota_{Y,X} : R[Y] \rightarrow \iota_{Y,X}(R[Y])$ is a ring isomorphism onto its image, it follows that the subring

$$S := \iota_{Y,X}(R[Y])$$

is also a unique factorization domain.

Since $p \in S$ and S is a unique factorization domain, there exist irreducible elements

$$q_1, \dots, q_n \in S$$

such that

$$p = q_1 \cdots q_n$$

in S , and this factorization is unique up to order and associates in S .

We now claim that each q_i is irreducible in $R[X]$. Indeed, by the lemma asserting that S is factorally closed in $R[X]$, together with the unit condition

$$S^\times = R[X]^\times \cap S,$$

Lemma 3.8 implies that every element irreducible in S is irreducible in $R[X]$. Hence

$$p = q_1 \cdots q_n$$

is a factorization of p into irreducibles in $R[X]$. This proves existence.

It remains to prove uniqueness.

Suppose

$$p = r_1 \cdots r_m = s_1 \cdots s_n$$

are two factorizations of p into irreducibles in $R[X]$.

Each factor r_i and each factor s_j is a polynomial involving only finitely many variables. Therefore there exists a finite subset $Z \subseteq X$ containing every variable that occurs in any of the polynomials

$$r_1, \dots, r_m, s_1, \dots, s_n.$$

Hence

$$r_i, s_j \in \iota_{Z,X}(R[Z]) \quad \text{for all } i, j.$$

Set

$$T := \iota_{Z,X}(R[Z]).$$

Since Z is finite, $R[Z]$ is a unique factorization domain, and therefore T is a unique factorization domain as well.

Now each r_i and each s_j is irreducible in $R[X]$. We claim that each is also irreducible in T . Indeed, if for example

$$r_i = ab$$

in T , then this is also a factorization in $R[X]$. Since r_i is irreducible in $R[X]$, one of a or b is a unit in $R[X]$. Because

$$T^\times = R[X]^\times \cap T,$$

that factor is also a unit in T . Thus r_i is irreducible in T . The same argument applies to each s_j . Therefore both

$$p = r_1 \cdots r_m \quad \text{and} \quad p = s_1 \cdots s_n$$

are factorizations of p into irreducibles in the unique factorization domain T . By uniqueness in T , it follows that

$$m = n$$

and, after reordering, r_i is associate to s_i in T for each i . Since association in T implies association in $R[X]$, the two factorizations are also the same up to order and associates in $R[X]$.

Thus factorization in $R[X]$ is unique.

We have proved that every nonzero nonunit element of $R[X]$ factors into irreducibles, and that such factorizations are unique up to order and associates. Therefore $R[X]$ is a unique factorization domain. $\square$